

Quantum Singular Value Transformation

A unified framework for quantum algorithms

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CDT QI Spring School

May 26, 2026

Topics to cover

- Lec 1: Quantum singular value transform (Tue)
- Lec 2: Quantum query lower bound (Wed)
- Lec 3: Quantum fine-grained complexity (Fri)

This lecture

- 1 Block Encodings
- 2 Quantum Signal Processing
- 3 Quantum Singular Value Transformation
- 4 Applications
- 5 Wrap-up

Goal. Understand the framework that unifies the major quantum algorithms.

- **Block encoding** — the right input model for matrices on a quantum computer.
- **QSP** — a single qubit + alternating rotations realises any admissible polynomial.
- **QSVT** — lift QSP to matrices: act on all singular values simultaneously.
- **Applications** — Hamiltonian simulation over and HHL become choices of polynomial.

The unifying question

Given a state $|\psi\rangle$ and access to an operator A , how do we apply $f(A)$ to the state, query-efficiently?

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- **Grover.** f projects onto solutions.
- **HHL.** $f(x) = 1/x$.
- **Hamiltonian simulation.** $f(x) = e^{-itx}$.
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Answer: QSVT. Pick a polynomial P approximating f , apply it via a *block encoding* of A .

$\Rightarrow$ Quantum algorithm design reduces to *approximation theory*.

Block encodings: a uniform interface for matrix access

Definition. A unitary U on a ancilla + n system qubits is an (α, a, ε) -block encoding of A if

$$\|A - \alpha (\langle 0^a| \otimes \text{Id}) U (|0^a\rangle \otimes \text{Id})\| \leq \varepsilon.$$

$U \approx$

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Why this object.

- Standardises “access to a non-unitary matrix”.
- Closed under products, sums, and tensor products.
- Many natural input models give block encodings.

Building block encodings

Sparse-access oracles. For s -sparse A with $\|A\|_{\max} \leq 1$, oracles for nonzero locations and entries give an $(s, O(\log N), 0)$ -block encoding using $O(1)$ oracle queries (Childs '10; Berry–Childs–Kothari '15).

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Linear combination of unitaries (LCU). If $A = \sum_{k=1}^L \alpha_k U_k$ with $\alpha_k > 0$, $\alpha := \sum_k \alpha_k$:

$$\text{PREP} |0\rangle = \frac{1}{\sqrt{\alpha}} \sum_k \sqrt{\alpha_k} |k\rangle, \quad \text{SEL} = \sum_k |k\rangle\langle k| \otimes U_k.$$

Then $(\text{PREP}^\dagger \otimes \text{Id}) \text{SEL} (\text{PREP} \otimes \text{Id})$ is an $(\alpha, \lceil \log L \rceil, 0)$ -block encoding.

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Hamiltonian from Pauli decomposition. $H = \sum_j \alpha_j P_j$ is a direct LCU, block-encoded with $\alpha = \sum_j |\alpha_j|$.

Quantum signal processing: a single qubit, alternating rotations

Signal unitary. For $x \in [-1, 1]$,

$$W(x) = \begin{pmatrix} x & i\sqrt{1-x^2} \\ i\sqrt{1-x^2} & x \end{pmatrix} = e^{i \arccos(x) X}.$$

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$$U_{\vec{\phi}}(x) = D(\phi_0) \prod_{k=1}^d (W(x) D(\phi_k)).$$

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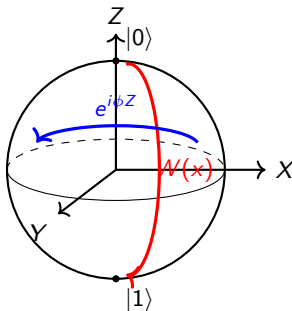
$$U_{\vec{\phi}}(x) = D(\phi_0) \prod_{k=1}^d (W(x) D(\phi_k)).$$

Lemma. The top-left entry of $U_{\vec{\phi}}(x)$ is a polynomial $P(x) \in \mathbb{C}[x]$ of degree $\leq d$ with $|P(x)| \leq 1$ on $[-1, 1]$.

d phase rotations interleaved with d signal calls $\Rightarrow$ degree- d polynomial.

QSP on the Bloch sphere

$W(x)$ rotates around X by $-2\arccos(x)$. $D(\phi)$ rotates around Z by -2ϕ .



Alternating X - and Z -rotations on a single qubit; $\langle 0| U_{\vec{\phi}}(x) |0\rangle$ encodes the resulting polynomial.

QSP achievability (Low–Chuang '17; GSLW '19)

Theorem. Any $P \in \mathbb{R}[x]$ of degree $\leq d$, definite parity, $|P| \leq 1$ on $[-1, 1]$ is realised by some $\vec{\phi}$ of length $d + 1$.

Phase computation. Given P , the phases $\vec{\phi}$ are computed classically in $\text{poly}(d)$ (Haah '19, Dong–Meng–Whaley–Lin '21).

Lifting QSP to matrices: QSVT (GSLW '19)

Let U be an $(\alpha, a, 0)$ -block encoding of A , with SVD

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Theorem (Gilyén–Su–Low–Wiebe '19). For real P of definite parity, degree d , $|P| \leq 1$ on $[-1, 1]$, there is a circuit using d uses of $U, U^\dagger$ and $O(d(a+1))$ extra gates that block-encodes

$$P^{\text{SV}}(A/\alpha) := \begin{cases} \sum_k P(\sigma_k) |u_k\rangle\langle v_k| & d \text{ odd,} \\ \sum_k P(\sigma_k) |v_k\rangle\langle v_k| & d \text{ even.} \end{cases}$$

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In words. What a degree- d polynomial does to a scalar in $[-1, 1]$, QSVT does to every singular value of a block-encoded matrix, using d oracle calls.

QSVT circuit

Let $\Pi = |0^a\rangle\langle 0^a| \otimes \text{Id}$. Define the projector-controlled phase

$$\Pi_\phi = e^{i\phi(2\Pi - \text{Id})}.$$

This applies phase $e^{i\phi}$ to the $\Pi = 1$ subspace and $e^{-i\phi}$ to the $\Pi = 0$ subspace (proof in appendix).

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$$U_{\vec{\phi}} = \Pi_{\phi_0} \prod_{k=1}^{d/2} (U^\dagger \Pi_{\phi_{2k-1}} U \Pi_{\phi_{2k}}).$$

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Why it works. The block-encoded operator A/α acts inside a family of 2D invariant subspaces, in each of which it works like the QSP.

Qubitization step 1: what U does to an eigenvector

Hermitian case. We prove the result for Hermitian A (eigendecomposition), which covers all applications in this lecture.

The general SVD case uses the Cosine-Sine decomposition (GSLW '19).

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Embed: $|\tilde{v}_k\rangle := |0^a\rangle |v_k\rangle$.

By the block-encoding definition, $(\langle 0^a| \otimes \text{Id}) U (|0^a\rangle \otimes \text{Id}) = A/\alpha$, so the $|0^a\rangle$ -component of $U|\tilde{v}_k\rangle$ is $\lambda_k |v_k\rangle$.

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Since U is unitary, the remainder has norm $\sqrt{1 - \lambda_k^2}$ and lies outside the $|0^a\rangle$ subspace:

$$U|\tilde{v}_k\rangle = \lambda_k |\tilde{v}_k\rangle + \sqrt{1 - \lambda_k^2} |\perp_k\rangle, \quad \Pi |\perp_k\rangle = 0.$$

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Π_ϕ in this basis.

$$\Pi_\phi|_{\mathcal{H}_k} = \begin{pmatrix} e^{i\phi} & 0 \\ 0 & e^{-i\phi} \end{pmatrix} = D(\phi), \quad \text{exactly the QSP phase rotation.}$$

$\Rightarrow$ Inside each $\mathcal{H}_k$, the QSVT circuit is a QSP sequence with signal λ_k .

QSVT = QSP run in parallel

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Consequence. Applying the full circuit $U_{\vec{\phi}}$:

- On $|\tilde{v}_k\rangle$: produces $P(\lambda_k) |\tilde{v}_k\rangle + (\text{orthogonal junk})$.
- On any input $|0^a\rangle |\psi\rangle$ with $|\psi\rangle = \sum_k \beta_k |v_k\rangle$: produces $|0^a\rangle \sum_k \beta_k P(\lambda_k) |v_k\rangle$ in the encoded block, plus garbage on orthogonal ancilla states.

Thus $U_{\vec{\phi}}$ is a block encoding of $P(A/\alpha) = \sum_k P(\lambda_k) |v_k\rangle\langle v_k|$. $\square$

Polynomial approximation is all you need

Three workhorses:

- **Sign function.** $\exists P$ of degree $O(\delta^{-1} \log(1/\varepsilon))$ with $|P(x) - \text{sgn}(x)| \leq \varepsilon$ on $[-1, -\delta] \cup [\delta, 1]$.

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$$\cos(tx) = J_0(t) + 2 \sum_{k \geq 1} (-1)^k J_{2k}(t) T_{2k}(x),$$

truncate at $d = O(|t| + \log(1/\varepsilon)/\log(\log(1/\varepsilon)/|t|))$.

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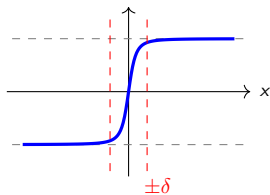
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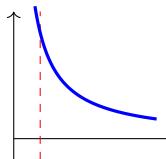
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Each function: smooth $\Rightarrow$ exponentially decaying Chebyshev coefficients $\Rightarrow$ low-degree polynomial approximation.

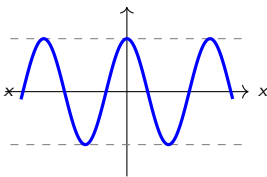
Polynomial gallery



Smoothed $\text{sgn}(x)$



$1/\kappa$
 $1/x$ on $[\kappa^{-1}, 1]$



$\cos(tx)$

- $\text{sgn} \rightarrow$ amplitude amplification & eigenvalue threshold.
- $1/x \rightarrow$ matrix inversion (HHL).
- $\cos / \sin \rightarrow$ Hamiltonian simulation.

Application I: Hamiltonian simulation — the idea

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Which polynomial? Use Euler's formula: $e^{-ixt} = \cos(xt) - i \sin(xt)$. Both $\cos(xt)$ and $\sin(xt)$ are well-approximated by Jacobi–Anger (Chebyshev) expansions of degree $d = O(\alpha t + \log(1/\varepsilon))$.

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Why two polynomials? QSVT applies *real* polynomials with definite parity. $\cos$ is even, and $\sin$ is odd — each is handled by a separate QSVT circuit, then combined via LCU: $e^{-iHt} = \cos(Ht) - i \sin(Ht)$.

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- 4 LCU combination: $e^{-iHt} = \cos(Ht) - i \sin(Ht)$.

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Setup. H Hermitian, $\|H\| \leq 1$, $(\alpha, a, 0)$ -block encoding U_H .

Recipe.

- 1 Approximate $\cos(\alpha tx)$ and $\sin(\alpha tx)$ by Jacobi–Anger truncations of degree $d = O(\alpha t + \log(1/\varepsilon))$.
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Optimality. Matches the no-fast-forwarding lower bound: $\Omega(\alpha t + \log(1/\varepsilon)/\log \log(1/\varepsilon))$ (Berry–Ahokas–Cleve–Sanders '07). The cost is *linear* in time t and *logarithmic* in precision $1/\varepsilon$.

Application II: matrix inversion — the idea

Problem. Given A with eigenvalues $\lambda_k \in [\kappa^{-1}, 1]$ and a state $|b\rangle = \sum_k \beta_k |v_k\rangle$, prepare $|x\rangle \propto A^{-1} |b\rangle = \sum_k \frac{\beta_k}{\lambda_k} |v_k\rangle$.

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Key insight. Matrix inversion = applying $f(x) = 1/x$ to each eigenvalue. QSVT can apply any polynomial to eigenvalues — so approximate $1/x$ by a polynomial!

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What QSVT gives us. Apply P^{SV} to the block-encoded A :

$$A = \sum_k \lambda_k |v_k\rangle\langle v_k| \xrightarrow{\text{QSVT}} P^{\text{SV}}(A) = \sum_k \frac{1}{2\kappa\lambda_k} |v_k\rangle\langle v_k| = \frac{1}{2\kappa} A^{-1}.$$

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Caveats.

- State prep for $|b\rangle$ can dominate the cost.
- Read-out: you get $|x\rangle \propto A^{-1}|b\rangle$ as a quantum state. Extracting a single entry costs $O(1)$ measurements, but reading all N entries costs $O(N)$ — no exponential speedup there.
- Dequantisation: for low-rank A , classical algorithms match this in $\text{poly}(\kappa)$ (Tang '19).

Why QSVT is the good framework

Unification table.

Algorithm	Target function	Degree	Prior framework
Amplitude amplification	$\text{sgn}(x)$ (threshold at $\sqrt{a}$)	$O(1/\sqrt{a})$	Grover '96
Hamiltonian simulation	$e^{-itx} = \cos(tx) - i \sin(tx)$	$O(\alpha t + \log \frac{1}{\epsilon})$	Product formulae
Matrix inversion	$1/x$ on $[\kappa^{-1}, 1]$	$O(\kappa \log \frac{\kappa}{\epsilon})$	HHL '09
Quantum walk search	$\text{sgn}(x)$ (threshold at $\sqrt{\delta}$)	$O(1/\sqrt{\delta})$	Szegedy '04

Optimality. Each application matches its known query lower bound (BBBV; BACS '07; HHL hardness).

If your task is “apply $f(A)$ to $|\psi\rangle$ ” and f admits a low-degree polynomial approximation, you have a quantum algorithm.

Take-aways

- **Block encoding** is the right input model for non-unitary operators.
- **QSP** realises any admissible polynomial of a ± 1 -bounded scalar via d alternating rotations of a single qubit.
- **QSVT** lifts QSP to act on all singular values of a block-encoded matrix simultaneously. The mechanism is the 2D invariant-subspace decomposition (“qubitization”).
- All of Grover, HHL, Hamiltonian simulation, and amplitude estimation become a *choice of polynomial*.
- Cost is dictated by approximation theory
- The resulting algorithms are typically query-optimal. (See next lecture)

Further reading

- Gilyén, Su, Low, Wiebe. *Quantum singular value transformation and beyond*. STOC '19.
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- Haah. *Product decomposition of periodic functions in quantum signal processing*. Quantum '19.

Questions?